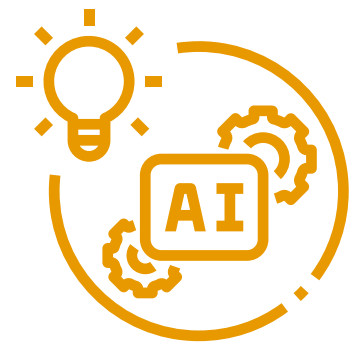


AGENTIC AI

CASE STUDY

leveraging **Generative AI**



OVERVIEW

A leading manufacturing company with high-volume production lines, seeking to improve operational efficiency by preventing equipment failures and minimizing downtime.

The client's goal was to deploy a Predictive Analytics solution, powered by Generative AI, to address critical pain points in their manufacturing processes. They needed a system capable of:

1. Predicting anomalies in equipment or process behavior before failures occur.
2. Automating root cause analysis (RCA) to identify why anomalies happen.
3. Delivering notifications and actionable next steps to the relevant teams.
4. Providing a customizable workflow for operators to tune the outputs for more accurate and contextually relevant suggestions.

InfoObjects suggested approach to build an LLM-based analytics system supplemented by a RAG pipeline for the failure cause / Correlation analysis of client's proprietary manufacturing data. To improve model accuracy, an RLHF feedback loop flow was implemented

“Implementing the Generative AI solution has been a game-changer for our manufacturing operations. The ability to predict anomalies in real-time and instantly provide root cause analysis has drastically reduced our downtime and improved overall efficiency. What used to take our operators over 20 minutes, now happens in seconds, which has allowed us to maintain continuous production and significantly cut operational costs.”

Director of Engineering
Manufacturing

APPROACH

- **Large Language Model (LLM) Workflow:** To predict equipment or process anomalies and extract relevant insights from the data.
- **Retrieval-Augmented Generation (RAG) Pipeline:** For root cause analysis, retrieving information from product manuals and historical records to identify the source of issues.
- **Reinforcement Learning with Human Feedback (RLHF):** To continuously improve model accuracy through feedback loops from operators.
- **DataBricks Unity Catalog & Model Serving:** For efficient data processing and governance.

Technology Stack

- Databricks Workflows
- Delta Live Table
- Databricks Autoloader
- Mosaic AI Vector Store
- Databricks hosted LLM (llama 3)
- ML Model for Anomaly Detection

RESULTS



62%
Faster
Issue Resolution



76%
Improved
Anomaly Detection



78%
Reduced
Manual effort

CONCLUSION

The proposed system leverages cutting-edge Databricks features and AWS Cloud integration to address the challenges of high-frequency sensor data management. By combining scalable data ingestion, advanced machine learning, and automated resolution workflows, this solution enhances operational efficiency, reduces downtime, and delivers immediate value.

**Contact us to setup your
FREE Generative AI Advisory Session**



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